Beneath the Surface: Detecting Procurement Cartels through Worker-Flow Networks*

Sergio Galletta[1], Darcio Genicolo-Martins[2], Tommaso Giommoni[3]

[1] *ETH Zürich*
[2] *INSPER*
[3] *University of Amsterdam*

Abstract

Brazil's antitrust authority secured administrative convictions against 35 firms in seven procurement cartels operating in the state of São Paulo between 2009 and 2019. We match these firms and the product markets delineated in the case files to 3.8 million bids on the state's electronic procurement platform and to the universe of matched employer–employee records, assembling the first conviction-anchored benchmark that integrates auction behavior, labor microdata, and antitrust rulings.

On a unified benchmark of roughly 2,300 close-margin pregão auctions (absolute bid margin $|MV| \leq 0.10$) in which at least one convicted firm participated during the active period of its cartel, canonical variance-based screens fire in the direction predicted by the bid-rotation model—bid dispersion and the coefficient of variation are weakly compressed (Cohen's $d \approx -0.07$ to -0.10) and the number of active bidders is modestly lower ($d \approx -0.24$)—yet the corresponding areas under the ROC curve (0.52–0.56) remain far too low for investigative triage, indicating near-total distributional overlap between cartel and clean auctions. A close-margin regression discontinuity in the spirit of Kawai et al. (2023) likewise yields a null. By contrast, network screens constructed from the same benchmark achieve markedly higher discrimination: inter-firm worker flows from the matched employer–employee data yield AUCs of 0.75 (shared workers) and 0.74 (Jaccard similarity); a co-bidding density screen yields 0.70. The worker-flow signal survives conditioning on sector and on co-bidding intensity. A specialization placebo restricted to non-cartel firm pairs bidding on the same product code in the same year

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Email addresses: `sergio.galletta@gess.ethz.ch` (Sergio Galletta), `darciogm1@insper.edu.br` (Darcio Genicolo-Martins), `t.giommoni@uva.nl` (Tommaso Giommoni)

absorbs approximately half of the worker-flow signal, leaving a cartel-specific residual—consistent with cartels both inheriting and reinforcing the personnel flows natural to thin specialized-labor markets.

These findings support a view in which collusion leaves institution-specific signatures: in markets with thin specialized-labor pools, the cheapest coordination technology operates through personnel flows rather than bid-pattern manipulation. For competition authorities with access to employer–employee microdata, worker-flow screens offer a deployable complement to canonical bid-pattern tests.

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1. Introduction

Do the canonical tools for detecting procurement cartels work when they are deployed outside the thick-market, high-enforcement environments in which they were developed? We assemble the first conviction-anchored detection benchmark that integrates auction-level bid data, matched employer–employee records, and formal antitrust rulings—linking seven convicted cartels in São Paulo to 3.8 million bids and the universe of formal-sector employment relationships. On this benchmark, classical variance-based screens and a close-margin regression discontinuity design yield AUCs of 0.52–0.56: directionally correct but operationally useless. By contrast, screens constructed from inter-firm worker flows in the employer–employee data achieve AUCs of 0.75—and the gap widens sharply in thin labor markets where collusion coordinates through personnel rather than bid patterns. This paper documents these facts, proposes a mechanism linking screen effectiveness to institutional primitives, and draws out the implications for enforcement design.

A growing literature uses statistical regularities in bidding behavior to infer collusion that is never documented directly in the administrative record. Classical screens exploit the compression induced by bid-rotation agreements: colluding firms submit prearranged prices that cluster tightly around the designated winner, yielding low dispersion, few active bidders, and predictable rotation patterns (Porter and Zona, 1993; Pesendorfer, 2000; Bajari and Ye, 2003; Conley and Decarolis, 2016; Imhof et al., 2018). A complementary and influential approach, recently advanced by Kawai et al. (2023), focuses on the winning–losing threshold: under competition, the observable characteristics of narrowly winning and narrowly losing firms should be continuous at a bid margin of zero, and any statistically sharp discontinuity at the cutoff identifies an anti-competitive wedge. Both approaches have been validated primarily against cases in high-income economies with deep formal labor markets, dense bidder pools, and strong institutional capacity for antitrust enforcement; their extension to thinner institutional environments rests on a small but growing empirical base. Wachs and Kertész (2019) apply a co-bidding network approach to roughly 150,000 Georgian procurement notices and show that group-level cohesion and exclusivity flag candidate cartels in the absence of criminal ground truth. Signor et al. (2020) exploit the dataset released by the Brazilian Federal Police's *Operação Lava Jato* investigation to build a probabilistic detector of collusive tenders in the Petrobras construction-cartel case, reporting accuracy between 81 and 96 percent. Huber

and Imhof (2019) combine classical statistical screens with supervised machine-learning classifiers and correctly predict 84 percent of Swiss tenders out of sample, demonstrating the gains from combining bid-distribution screens with flexible functional forms. What has been missing is a direct test against an independent criminal ground truth in a setting with *both* structurally different institutional primitives *and* access to administrative employer–employee microdata. The São Paulo–CADE–RAIS linkage we build offers exactly that test, and is, to our knowledge, the first cartel-detection benchmark to integrate auction-level behavioral data, employer–employee matched labor records, and formal antitrust convictions. The policy stakes are first-order: middle-income competition authorities increasingly rely on off-the-shelf screens developed and validated abroad to triage investigations (Organisation for Economic Co-operation and Development, 2015; Bosio et al., 2022), and if those screens systematically mislabel cartels under local conditions, existing enforcement pipelines may be misfiring in precisely the settings where deterrence capacity is weakest.

Our primary evaluation sample is a *cartel-active* benchmark of roughly 2,300 close-margin pregão auctions in which at least one convicted firm participated during the active period of its cartel (details of sample construction in Section 5).

Our first set of results documents the limited discriminative power of canonical screens in this institutional setting. On the unified cartel-active benchmark of roughly 2,300 close-margin pregão auctions ($|MV| \leq 0.10$), classical variance screens fire in the direction predicted by the bid-rotation model: bid dispersion and the coefficient of variation are weakly compressed at cartel-tainted auctions relative to clean controls (Cohen's $d \approx -0.07$ to -0.10), and the number of active bidders is modestly lower ($d \approx -0.24$), with Welch t-statistics that are statistically significant given the large control sample. Yet the effect sizes are small and the distributional overlap between cartel and clean auctions is near-total: the areas under the receiver-operating-characteristic curves (AUCs) for these screens range from 0.52 (coefficient of variation) to 0.56 (bidder count)—above chance but far too low for investigative triage. A Wilcoxon rank-sum test confirms the weak rank-order separation (rank-sum $p < 10^{-4}$ for each, aligning with the AUC direction). Classical screens are therefore directionally consistent with theory but operationally inadequate as cartel detectors in this setting: a triage tool built on these statistics would generate an unacceptable rate of false positives and false negatives relative to the conviction-anchored ground truth. The same qualitative pattern holds on the strict pair-matched sample of 14 auctions—canonical direction, low discriminative

power, and mostly non-significant individual t-tests owing to small-N constraints. A direct replication of the Kawai close-margin RD on the cartel-active subsample yields no statistically detectable discontinuity in incumbency or last-bid behavior at the cutoff, and a density test in the spirit of Imhof et al. (2018) detects no manipulation of the running variable. The anytime-listed sensitivity sample of 107 auctions does produce positive Δ estimates on variance screens with $p < 0.02$, but we interpret this as a mechanical consequence of the 93 out-of-window auctions in that sample—auctions in which firms that had been cartelists in another year are bidding against each other under competitive conditions—rather than as evidence that cartel auctions exhibit elevated dispersion. In short, while canonical variance-based screens exhibit weak directional signal, neither they nor the close-margin RD of Kawai et al. (2023) achieve the discriminative power required to operationally detect the cartels that CADE has formally convicted.

Our second set of results shows that network-based screens constructed from administrative labor data recover strong and statistically robust cartel detection on the cartel-active benchmark. For each auction, we construct two network screens from the five-year window of RAIS records preceding the tender: the number of workers who have moved between the winner and the runner-up firms, and the Jaccard similarity of the two firms' historical workforces. On the cartel-active sample of roughly 2,300 close-margin auctions, these screens yield AUCs of 0.750 and 0.739 respectively—outperforming the best classical screen by roughly 20 percentage points and representing, to our knowledge, the first robust AUCs reported for procurement cartel detection against a formal criminal ground truth. A third screen, built from the co-bidding network of shared tender participations, yields an AUC of 0.695, providing independent corroboration through a distinct information channel. Critically, the worker-flow signal is neither an artifact of sectoral composition nor a mechanical re-expression of co-bidding intensity. Restricting attention to firm pairs that share the same 2-digit sectoral classification reduces the worker-flow AUC only marginally, from 0.739 to 0.730, and the residual of worker-flow after partialling out the log co-bidding density of each pair achieves an AUC of 0.749—essentially unchanged from the raw baseline, indicating that worker mobility carries detection information above and beyond what co-bidding patterns alone can reveal. A more demanding test separates cartel-specific coordination from passive same-market labor circulation: we construct a *specialization placebo* of non-cartel firm pairs bidding on the same BEC item code in the same year as each cartel-active auction, and compare worker-flow distributions across cartel, placebo, and global clean samples. Cartel

pairs share at least one worker in 57.9% of cases, compared with 31.1% for same-market placebo pairs and 9.9% for pairs in other markets. In a decomposition of the worker-flow AUC gain relative to the global clean baseline, roughly half is absorbed by the same-market specialization component and roughly half remains as a cartel-specific residual. Both components are consistent with our institutional-thinness mechanism: cartels inherit the labor-circulation patterns natural to thin specialized-labor markets and additionally reinforce those flows to sustain coordination. On the strict pair-matched sample ($N = 10$ after close-margin filtering), worker-flow AUCs decline to near chance; we attribute this to power constraints at small sample sizes rather than to any breakdown in the underlying signal, and we emphasize that the cartel-active benchmark is where the discriminative power of any screen—classical or network-based—can be meaningfully evaluated. The conviction-anchored benchmark does not merely declare cartels detectable in São Paulo; it pinpoints the channel through which the detection signal propagates—worker mobility, not bid dispersion.

We interpret these findings through the lens of institutional thinness. In settings where the universe of firms capable of bidding on a given product is small and the formal-sector labor market for the specialized workers that serve those bids is shallow, coordination among colluding firms generates naturally detectable traces in worker flows: personnel rotations, key-employee transfers, and shared specialized labor become vehicles for, and therefore signatures of, the collusive communication that substitutes for the statistical-pattern coordination captured by canonical screens in thicker markets. This mechanism—which we develop formally in a simple model in Section 4—rationalizes both the low discriminative power of bid-dispersion screens and the success of network-based screens in the Brazilian setting. Critically, it also predicts the two-component decomposition of the worker-flow signal documented by our specialization placebo: because thin markets generate spontaneous labor circulation among same-market firms whether or not they are colluding, a non-cartel placebo restricted to same-market firm pairs should already exhibit elevated worker-flow overlap relative to random pairs; and cartel pairs should exhibit still greater overlap on top of that baseline. Both predictions are borne out in the data. The mechanism further predicts that classical screen-based enforcement strategies will systematically under-detect cartels in precisely the institutional environments where their deterrence value would be highest.

This paper makes three contributions. *First*, conceptually, we advance the view that collusion leaves *institution-specific* signatures: the effectiveness of a detection screen de-

pends on which coordination technology is cheapest for cartel members to operate in the relevant institutional environment. In thick labor markets, collusion coordinates through bid patterns and classical screens fire; in thin specialized-labor markets, coordination piggybacks on personnel flows that classical screens cannot see. We formalize this in a simple model (Section 4.6) and confirm its central prediction empirically: the gap between classical and worker-flow screens widens from 10 percentage points in thick markets to 33 percentage points in thin markets. This framing organizes and partially reconciles the growing patchwork of contradictory findings in the detection literature (Conley and Decarolis, 2016; Imhof et al., 2018; Kawai et al., 2023). *Second*, substantively, we introduce conviction-anchored detection benchmarks by linking CADE rulings, BEC bid-level records, and RAIS employer–employee data, and we show that worker-flow and co-bidding network screens constructed from these sources substantially outperform classical screens in detecting convicted cartels. Because such data are increasingly available to national competition authorities in middle-income economies, the screens we propose are directly deployable at negligible marginal cost. *Third*, methodologically, we deliver the first stress test of canonical cartel-detection frameworks against an independent criminal ground truth in a middle-income setting, documenting that classical variance-based screens (Porter and Zona, 1993; Bajari and Ye, 2003) and the close-margin RD of Kawai et al. (2023) yield AUCs of only 0.52–0.56—directionally correct but operationally inadequate—while network screens achieve 0.75.

The remainder of the paper is structured as follows. Section 3 describes the institutional setting of São Paulo's public procurement and the CADE cartel cases that anchor our ground truth. Section 2 situates our contribution in the procurement-collusion detection literature and in the adjacent literatures on worker mobility and law-and-economics enforcement. Section 5 describes the BEC, RAIS, and CADE data sources and the construction of the cartel-active and pair-matched benchmark samples. Section 4 presents our empirical strategy—the RD replication, the portfolio of classical screens, and the network-based screens—and sketches a simple theoretical framework linking screen predictions to institutional primitives. Section 6 reports the main results. Section 7 discusses policy implications and concludes.

2. Related Literature

This paper engages three distinct literatures: cartel detection via bid-pattern screens, network-based approaches to collusion identification, and the economics of inter-firm worker mobility.

2.1. Detection through bid-pattern screens

A foundational literature exploits the prediction that bid-rotation agreements compress the statistical properties of bids. Porter and Zona (1993) introduced variance-based screens to the railroad freight cartel of the 1880s, showing that collusive phases exhibit lower bid dispersion than competitive phases when the cartel's internal records provide an exogenous regime switch. Pesendorfer (2000) documented bid rotation in Florida school-milk procurement, demonstrating that cartel members systematically coordinate winning turns. Bajari and Ye (2003) proposed exchangeability tests for bidder costs in sealed-bid auctions, applying them to highway paving contracts in the U.S. Midwest; rejections of exchangeability flag candidate collusive pairs. Conley and Decarolis (2016) extended these ideas to Italian procurement, using pair-level bid-correlation tests to detect bidder groups. Imhof et al. (2018) combined classical variance and bid-level screens with density tests in a Swiss procurement setting, demonstrating that multi-screen portfolios improve detection relative to any single indicator. Abrantes-Metz et al. (2006) provided practical guidance on implementing price-fixing screens, cautioning against single-indicator approaches. Harrington (2008) synthesized the design principles for detection mechanisms, emphasizing that the informativeness of any screen depends on the competitive counterfactual and the cartel's coordination technology—a point our findings make concrete.

More recently, Kawai et al. (2023) proposed a regression discontinuity design centered on the winning–losing margin in Japanese construction procurement: under competition, firm characteristics should be continuous at the bid margin of zero, and any sharp discontinuity identifies a non-competitive wedge. Their approach has high internal validity but, as our results show, may lack discriminative power in settings where collusion does not concentrate at the bid margin. Chassang and Ortner (2019) offered a complementary theoretical framework in which bid constraints endogenously shape the collusive technology, predicting that auction-format-specific screens may outperform generic variance

tests—a prediction consistent with our finding that classical screens behave differently across pregão and convite formats.

A common thread in this literature is that almost all validation exercises rely on ground truth from high-income economies with deep bidder pools and strong enforcement capacity. Levenstein and Suslow (2006) documented the institutional determinants of cartel success and duration across 81 international cartels, noting that market structure, transparency, and enforcement intensity jointly shape collusive outcomes. What remains scarce is direct validation of off-the-shelf screens against criminal ground truth in middle-income settings with structurally different labor markets and thinner bidder pools—precisely the gap our conviction-anchored benchmark addresses.

2.2. Network-based and machine-learning approaches

A growing literature moves beyond bid distributions to exploit relational and network information. Wachs and Kertész (2019) applied co-bidding network analysis to roughly 150,000 Georgian procurement notices, showing that group-level cohesion and exclusivity metrics flag candidate cartels even in the absence of criminal ground truth. Signor et al. (2020) exploited the dataset released by Brazil's *Operação Lava Jato* investigation to build a probabilistic detector of collusive tenders in the Petrobras construction-cartel case, reporting accuracy between 81 and 96 percent. Huber and Imhof (2019) combined classical statistical screens with supervised machine-learning classifiers and correctly predicted 84 percent of Swiss tenders out of sample, demonstrating the gains from combining bid-distribution features with flexible functional forms. Clark et al. (2018) studied collusion and corruption jointly in Québec construction procurement, documenting how collusive networks co-evolve with political connections and how entry deterrence interacts with bid rigging.

These network approaches share a common insight: collusion is fundamentally a relational phenomenon, and screens that exploit the structure of firm–firm interactions can extract signal that univariate bid statistics miss. Our contribution builds on this insight but shifts the network channel from co-bidding patterns (observable within the procurement data itself) to worker mobility across firms (observable only in matched employer–employee administrative records). This shift is consequential because worker-flow networks are constructed from an entirely independent data source, reducing the risk of circularity between the screen and the behavioral patterns it is designed to detect.

2.3. Worker mobility as an inter-firm transmission channel

A separate literature in labor economics establishes that worker flows between firms serve as a vehicle for knowledge transmission, coordination, and productivity spillovers. Stoyanov and Zubanov (2012) showed that productivity gains propagate across Danish firms through the mobility of workers, with the magnitude of the spillover proportional to the productivity gap between sending and receiving firms. Serafinelli (2019) documented similar patterns in the Veneto region of Italy, demonstrating that workers moving from high-productivity firms raise wages and productivity at destination firms. Jarosch et al. (2021) developed an equilibrium model of learning from coworkers, showing that team composition and worker reallocation jointly determine firm-level human capital accumulation. Together, these results establish that worker flows are not merely a labor-market phenomenon but an active channel of inter-firm information transmission.

We bring this insight to the collusion-detection setting. If worker mobility transmits knowledge and facilitates coordination between firms in general, it may also serve as a coordination technology for cartel members—and, critically, leave detectable traces in administrative records. In markets with thin specialized-labor pools, where the universe of workers with the requisite expertise is small, the signal-to-noise ratio of worker-flow screens should be highest: cartel-related personnel transfers stand out against a sparse baseline of spontaneous mobility. Our specialization placebo formalizes this intuition by decomposing the worker-flow detection signal into a same-market specialization component (present even among non-colluding firms in thin markets) and a cartel-specific residual (the incremental coordination signature). To our knowledge, ours is the first paper to deploy matched employer–employee data as a cartel-detection screen and to validate it against formal antitrust convictions.

3. Institutional Background

3.1. Public procurement in São Paulo: the BEC platform

Public procurement in the state of São Paulo is conducted through a decentralized system of purchasing agencies, each referred to as a Public Buyer Unit (PBU). Since 2007, the use of the state's electronic procurement platform, the *Bolsa Eletrônica de Compras* (BEC), has been mandatory for PBUs when acquiring common goods and services. The BEC platform centralizes tender procedures and records detailed information on each

procurement process. It plays a crucial role in São Paulo's procurement operations: for instance, in 2018 alone, approximately US$1.7 billion worth of goods and services were procured via BEC. Over the period from 2009 onward, the BEC system handled more than R$7.4 billion (approximately 34% of all state government purchases) for the State Health Secretariat (SES/SP) and other agencies in São Paulo. Given the large scale and value of these tenders, ensuring fairness and preventing collusion in the procurement process is of paramount importance.

The BEC catalog contains a comprehensive listing of standardized items and services, classified by category. Nearly 55,000 purchase offers were issued through BEC during our study period, covering about 6,350 distinct items ranging from medical supplies to office equipment. Each purchase offer (PO) is a digital solicitation document that identifies one or more items to be procured, along with the required quantities and specifications. Each item within a PO is auctioned separately; thus a unique procurement transaction can be defined by a combination of a PO number and an item code (POI). The BEC records include rich information for each POI, including the reference (reserve) price set by the buyer, the number of participating firms, all bids submitted (identifying winners and losers), the auction format used, and the identities of the buyer (PBU) and auctioneer.

São Paulo's PBUs predominantly use two types of competitive tendering procedures available on BEC: (i) a sealed-bid tender (*convite*) and (ii) a multi-round descending auction (*pregão eletrônico*). In a sealed-bid tender, firms submit confidential bids by a specified deadline. After the submission period closes, all bids are opened and the contract is awarded to the qualified bidder with the lowest bid price. The sealed-bid format is typically used for purchases up to a value limit of R$176,000 (approximately US$35,000). In contrast, the descending auction (pregão) has no value limit and involves an open, interactive bidding phase. Initially, firms submit sealed proposals which are evaluated for qualification. Then, all qualified bids are announced (with the bidders' identities concealed) and an open auction phase commences. During this phase, which lasts for an initial 20 minutes, bidders can submit progressively lower bids, knowing the current lowest bid in real time. If a new bid is placed in the final minutes of this period (between 16 and 20 minutes), the auction is automatically extended by an additional 4 minutes. The auction concludes when no new bids arrive for 4 consecutive minutes. The winner is the bidder offering the lowest price at the end of this process, provided that price is below the government's preset reference price.

The key difference between the two formats lies in the opportunity for iterative bid-

ding. The pregão format allows real-time bid competition and, after determining the lowest bid through the auction, a brief negotiation or confirmation step to potentially further reduce the price. In contrast, the convite is a single-shot process with no negotiation stage. From an oversight perspective, the open auction format provides more transparency during the bidding process (as all bids are observable in real time, albeit anonymously), whereas the sealed-bid format concentrates all transparency at the moment of opening. Both mechanisms are subject to anti-collusion regulations and audit processes, but their procedural differences can influence how collusion or favoritism might manifest. Our primary analysis focuses on pregão auctions, which account for the majority of BEC procurement by value and for which the close-margin regression-discontinuity framework is most naturally motivated; we report results on convite as a complementary robustness check.

3.2. Antitrust enforcement: CADE and the conviction-anchored ground truth

Brazil's federal competition authority, the *Conselho Administrativo de Defesa Econômica* (CADE), is an autonomous agency within the Ministry of Justice responsible for investigating and adjudicating anticompetitive conduct. CADE's Tribunal renders administrative decisions (not criminal convictions in the judicial sense) that carry binding fines and behavioral remedies. Its investigative arm, the *Superintendência-Geral* (SG), conducts preliminary inquiries and administrative proceedings, often drawing on leniency agreements (*Termos de Compromisso de Cessação*, TCCs) in which cartel participants cooperate in exchange for reduced penalties. The typical enforcement pipeline proceeds from an initial complaint or leniency application, through a formal administrative proceeding, to a Tribunal decision that either convicts or acquits the named firms.

Our ground truth consists of CADE administrative proceedings in which the Tribunal found firms guilty of cartel conduct in product markets relevant to procurement in the state of São Paulo. We identified these cases through a systematic search of CADE's public case repository, restricting attention to proceedings with final Tribunal decisions (not pending or preliminary-stage cases) that specify a cartel period and list named firms with identifiable CNPJs (the Brazilian taxpayer identification number for legal entities). The resulting set contains seven distinct cartels operating in São Paulo procurement markets between 2004 and 2019, spanning sectors as diverse as pharmaceuticals, school meals, trash bags, solar water heaters, and metropolitan transit. Together, these cartels

involve 35 convicted firms whose 8-digit CNPJ roots can be matched to the BEC bidding records.

The critical feature of this ground truth for our detection exercise is that CADE's case files delineate both the *firms* and the *product-market-time windows* over which each ring was found to have operated. This allows us to label individual BEC auctions as cartel-tainted—not merely firms as cartel-affiliated—by matching each convicted firm's CNPJ root and the CADE-specified conduct period to the corresponding BEC tenders by item code and year. The resulting conviction-anchored benchmark thus incorporates both the identity and timing dimensions of collusion, rather than treating all auctions involving a flagged firm as equally suspicious.

4. Empirical Strategy

We deploy four complementary detection approaches on the conviction-anchored benchmark described in Section 5: a close-margin regression discontinuity (Section 4.1), classical bid-distribution screens (Section 4.2), network-based screens constructed from worker-flow and co-bidding data (Section 4.3), and a specialization placebo that decomposes the network signal into market-specialization and cartel-specific components (Section 4.5). We evaluate each screen's discriminative power using the area under the receiver-operating-characteristic curve (Section 4.4). Section 4.6 sketches a simple model linking the informativeness of each screen family to institutional primitives.

4.1. Close-margin regression discontinuity

Following Kawai et al. (2023), we exploit the winning–losing threshold in close-margin pregão auctions. Let MV_a denote the margin of victory in auction a, defined as the percentage difference between the runner-up bid and the winning bid:

$$MV_a \;=\; \frac{b_a^{\text{runner-up}} - b_a^{\text{win}}}{b_a^{\text{win}}}\,.$$

The running variable is MV_a, with a cutoff at zero: auctions with $MV_a \approx 0$ are those in which the winner barely underbid the closest rival. Under competitive conditions, the observable pre-determined characteristics of narrowly winning and narrowly losing firms should be continuous at $MV_a = 0$. Any statistically sharp discontinuity at the cutoff identifies a non-competitive wedge—either collusion or external favoritism.

We estimate the discontinuity using local-polynomial regression with the bandwidth selector of Imhof et al. (2018). The two primary outcomes are *incumbency*—an indicator equal to one if the firm won the immediately preceding auction in the same market—and *last bidder*—an indicator for whether the firm submitted the final bid in the auction. A positive jump in incumbency at $MV = 0$ is consistent with bid rotation (the designated winner is systematically a past winner); a jump in last-bidder status is consistent with strategic bid timing. We supplement the RD estimates with a density test on the running variable, following Imhof et al. (2018), to assess whether there is manipulation of the bid margin itself—a prerequisite for interpreting discontinuities causally.

4.2. Classical bid-distribution screens

We implement four classical screens computed at the auction level on firm-best bids (one bid per firm, retaining the firm's lowest bid when multiple rounds exist):

1. **Coefficient of variation** (CV_a): the standard deviation of firm-best bids divided by their mean. Under bid rotation with cover bidding, CV_a is predicted to be lower at cartel-tainted auctions because bids cluster around the designated winner's price (Porter and Zona, 1993; Abrantes-Metz et al., 2006).
2. **Bid spread** (S_a): the range of firm-best bids normalized by their mean, $S_a = (\max - \min)/\text{mean}$. Like CV_a, the spread is predicted to be compressed under cover bidding.
3. **Margin of victory** (MV_a): defined above. Under bid rotation, the winning margin is predicted to be *small* because the designated winner undercuts cover bids by only the minimum necessary to win.
4. **Number of active bidders** (N_a): the count of distinct firms submitting bids. Collusive entry deterrence or market-division agreements predict *fewer* bidders at cartel-tainted auctions (Marshall and Marx, 2012).

Each screen is evaluated on the cartel-active benchmark ($|MV| \leq 0.10$), comparing the distribution of cartel-tainted auctions to clean controls. We report the Welch t-statistic for the mean difference, Cohen's d for effect-size calibration, the Wilcoxon rank-sum p-value for rank-order separation, and the AUC (Section 4.4).

4.3. Network-based screens

We construct three screens from administrative data sources external to the bid records themselves.

Worker-flow screens. For each auction-level (winner, runner-up) firm pair, we measure inter-firm labor connectivity using the five-year window of RAIS employer–employee records preceding the tender year. Two variants capture different aspects of labor proximity:

- **Shared workers** (W_{ij}): the number of distinct workers (identified by their PIS number) who appear in the employment records of both firm i and firm j at any point during the five-year window. A higher count indicates denser personnel flows between the two firms.
- **Jaccard similarity** (J_{ij}): the ratio $J_{ij} = W_{ij}/(N_i + N_j - W_{ij})$, where N_i and N_j are the total number of distinct workers employed by each firm over the window. The Jaccard index normalizes the shared-worker count by the size of each firm's workforce, ensuring that the screen is not mechanically driven by firm size.

The hypothesis is that cartel coordination leaves detectable traces in personnel flows: managers, salespeople, or technical staff who rotate between colluding firms serve as informal communication channels (Stoyanov and Zubanov, 2012; Serafinelli, 2019). Under the institutional-thinness mechanism we develop below, the worker-flow signal should be strongest in markets with thin specialized-labor pools.

Co-bidding screen. For each (winner, runner-up) pair, we count the number of auctions in which both firms submitted bids over the full sample period (C_{ij}). Firms that co-bid frequently may be coordinating market presence. The co-bidding screen provides an independent network channel constructed entirely within the procurement data, serving as a comparator for the worker-flow screens, which are constructed from a wholly external source.

4.4. Detection evaluation: AUC

For each screen s, we evaluate discriminative power by computing the area under the receiver-operating-characteristic curve (AUC). The AUC is the probability that a randomly drawn cartel-tainted auction scores higher (more cartel-like) on screen s than a randomly drawn clean auction. For screens where lower values are predicted under collusion (CV, S, MV, N), we negate the scores before computing the AUC, so that a value above 0.5 always indicates detection in the predicted direction. An AUC of 0.5 corresponds to chance (no separation), and an AUC of 1.0 to perfect separation.

We compute the AUC via the Mann–Whitney U statistic with average-rank tie-handling:

$$\text{AUC} = \frac{U_1}{n_1 \times n_0},$$

where U_1 is the sum of ranks of cartel-tainted auctions minus $n_1(n_1+1)/2$, and n_1 and n_0 are the cartel and clean sample sizes, respectively. All screens—classical and network-based—are evaluated on the same unified close-margin benchmark ($|MV| \leq 0.10$), ensuring apples-to-apples comparability. Appendix A.1 reports AUCs at six alternative bandwidths (0.02–0.50); results are stable across the full range.

4.5. Specialization placebo

The worker-flow AUC may reflect two distinct channels: (i) *market specialization*—firms operating in the same thin product market share workers because the pool of specialized labor is small, regardless of whether they collude—and (ii) *cartel-specific coordination*—colluding firms engage in additional personnel transfers to sustain communication and monitoring. To disentangle these channels, we construct a three-way sample decomposition:

1. **Cartel**: auction-level (winner, runner-up) pairs in which at least one firm is cartel-active, drawn from the close-margin benchmark ($N \approx 2{,}300$ for pregão).
2. **Placebo (same market)**: non-cartel firm pairs bidding on the same BEC item code in the same year as each cartel-active auction. These pairs face the same product-market labor pool as cartel pairs but are not colluding (by the ground-truth definition).
3. **Clean (global)**: all remaining non-cartel pairs in the close-margin benchmark.

If the entire worker-flow signal were driven by market specialization, the placebo pairs should exhibit the same AUC as cartel pairs (relative to the global clean baseline). If instead there is a cartel-specific residual, cartel pairs should exhibit *higher* worker-flow overlap than same-market placebo pairs. In a linear decomposition of the total AUC gain (cartel vs. clean minus 0.5), we attribute the fraction absorbed by the placebo to market specialization and the remainder to cartel-specific coordination.

4.6. A simple model of institution-dependent coordination

We sketch a parsimonious framework to organize the empirical predictions. Consider a market with $n \geq 2$ firms that can form a cartel. The cartel must coordinate on a

winning bid for each auction. Two coordination technologies are available:

- **Bid-pattern coordination** (cost c_b): the cartel arranges bids so that cover bids cluster around the designated winner's price. This technology leaves traces in bid-distribution statistics (CV, spread, bidder counts) and is the mechanism assumed by classical screens.

- **Personnel-flow coordination** (cost $c_w(L)$, decreasing in the thickness L of the specialized-labor pool): the cartel coordinates through the transfer or rotation of key personnel—managers, salespeople, or technical staff—between member firms. This technology leaves traces in worker-flow statistics but not necessarily in bid distributions.

Each cartel chooses the cheapest available technology. In thick markets (large L), $c_w(L) > c_b$, and the cartel coordinates through bids: classical screens fire. In thin markets (small L), $c_w(L) < c_b$, and the cartel coordinates through personnel: worker-flow screens fire, but classical screens may not, because the bid distribution is not distorted by the coordination mechanism.

This framework generates three testable predictions:

1. **Direction**: In thin-labor markets, classical screen AUCs should be near chance; worker-flow AUCs should be above chance. The reverse holds in thick-labor markets.
2. **Placebo decomposition**: Because thin specialized-labor pools generate spontaneous inter-firm labor circulation even among non-colluding firms, a same-market placebo should exhibit elevated worker-flow overlap relative to random pairs (the specialization component). Cartel pairs should exhibit still greater overlap on top of that baseline (the coordination component). Both components should be present in thin-market data.
3. **Policy implication**: Enforcement strategies that rely exclusively on classical bid-pattern screens will systematically under-detect cartels in thin-labor environments, precisely where deterrence capacity may be weakest.

The model is deliberately stylized: it abstracts from dynamic considerations (repeated games, leniency incentives), from the possibility that firms use both technologies simultaneously, and from heterogeneity in c_b across auction formats. The framework

is offered not as a structural model to be estimated but as an organizing device for interpreting our empirical findings and for specifying the conditions under which the classical-vs.-network screen comparison is most informative.

5. Data

Our analysis draws on three administrative data sources: the BEC procurement platform (bid-level records), the RAIS employer–employee matched panel (worker-flow construction), and CADE administrative proceedings (cartel ground truth). This section describes each source and the construction of the conviction-anchored benchmark samples.

5.1. BEC: bid-level procurement data

We use the universe of bid-level records from the *Bolsa Eletrônica de Compras* (BEC) platform for the period January 2009 to December 2019. The data comprise 3,866,407 individual bidding transactions submitted by 19,007 distinct firms across 169,607 unique procurement items. Each record identifies the purchase offer (PO) and item code, the participating firm (by CNPJ), the bid amount, the auction outcome (win or loss), the auction format (pregão or convite), and the identity of the purchasing agency. We restrict attention to competitive tenders with at least one winner and one loser, excluding canceled tenders and single-bid auctions. We define a unique auction as a (PO number $\times$ item code) pair, henceforth POI.

For each POI, we construct the auction-level variables described in Section 4.2: the coefficient of variation, bid spread, margin of victory, and number of active bidders, computed on firm-best bids (retaining each firm's lowest bid when multiple rounds exist). We define markets following Kawai et al. (2023): two firms belong to the same market if they participated at least once in the same auction for the same item code. This transitive market definition yields 7,581 distinct markets.

5.2. RAIS: employer–employee matched panel

The *Relação Anual de Informações Sociais* (RAIS) is the universe of formal-sector employment relationships in Brazil, collected annually by the Ministry of Labor. Every establishment with a CNPJ must file a RAIS declaration listing each employment bond active at any point during the calendar year, identified by the worker's PIS/PASEP

number and the establishment's full 14-digit CNPJ. We use RAIS vinculos for the years 2009–2017, restricting to establishments whose 8-digit CNPJ root appears in the BEC firm universe (approximately 19,000 roots). The resulting panel contains all formal-sector employment bonds at BEC-active firms over the nine-year window.

From this panel, we construct the firm–firm worker-flow edge list described in Section 4.3. For each pair of BEC firms (i, j), we count the number of distinct PIS numbers that appear in the employment records of both firms at any point during the nine-year window (the shared-worker count W_{ij}) and compute the Jaccard similarity J_{ij}. We retain only edges with $W_{ij} \geq 1$; the remaining pairs are zero-imputed when joined to the auction-level benchmark.

5.3. CADE: cartel ground truth

The ground truth is constructed from CADE administrative proceedings with final Tribunal decisions convicting named firms of cartel conduct in São Paulo procurement markets (see Section 3 for institutional background). We identified seven relevant cartels spanning the sectors listed in Table 1, involving 35 firms with matchable CNPJ roots. For each cartel, the CADE case files specify a conduct period (start and end years) and the product market in which bid rigging or market division was found to have occurred.

We match each convicted firm's 8-digit CNPJ root to the BEC bidding records by firm identifier. We then label each BEC auction as *cartel-tainted* if at least one convicted firm participated during the CADE-specified conduct period for the relevant product market and year. This firm $\times$ period $\times$ product matching ensures that auctions are labeled as cartel-tainted only when a convicted firm was bidding during the window in which its cartel was found to be active—not merely because the firm appears in a CADE database at some other time.

5.4. Sample construction

We construct three conviction-anchored detection samples, all restricted to the close-margin envelope $|MV| \leq 0.10$:

1. **Cartel-active benchmark** ($N \approx 2{,}300$ for pregão; $N = 263$ for convite): auctions in which at least one convicted firm participated during the active period of its cartel. This is the primary evaluation sample on which all screens are tested.

Table 1: Convicted Cartels in São Paulo Procurement Markets

Sector	CADE process	Period	Firms	In BEC	Conduct type
Trens e metrôs	08700.004617	1999–2013	13	10	Bid rigging
Merenda escolar	08012.010022	2006–2013	7	6	Market division
Medicamentos	08012.002222	2007–2011	8	8	Hub-and-spoke
Sacos de lixo	08700.005789	2008–2014	3	3	Bid rigging / price fixing
Aquecedores solares	08012.001273	2009–2010	5	4	Bid rigging
Cafeterias Infraero	08700.007278	2014	6	3	Bid rigging
Transporte escolar	08700.005876	2019	2	2	Bid rigging
Total (unique CNPJ roots)			35	36	

Notes: "Firms" counts unique 8-digit CNPJ roots per cartel; "In BEC" counts those matched to BEC bidding records. Total is after deduplication across cartels. Conduct periods are hand-curated from CADE Tribunal decisions and press releases (see Appendix for full audit). Source: script 12.

2. **Strict pair-matched sample** ($N = 10$ for pregão after close-margin filtering): auctions in which *both* the winner and the runner-up are firms convicted in the same cartel during the same conduct period. This sample is the gold standard for detecting coordination between known cartel partners but is severely power-limited.
3. **Anytime-listed sensitivity sample** ($N \approx 80$ for pregão): auctions in which both the winner and runner-up appear in a CADE conviction for the same cartel process, but the auction year may fall outside the prosecuted conduct window. Of the 107 pre-filtering auctions in this sample, 93 are out-of-window and 100 come from a single cartel (the medicamentos hub-and-spoke case, CADE process 08012.002222/2011-09). Because treating out-of-window bidding by former cartelists as collusive behavior requires assumptions we do not attempt to resolve, this sample is reported only as a sensitivity check.

The *clean control* consists of all close-margin auctions in which no cartel-listed firm participated ($N = 527{,}259$ for pregão; $N = 519{,}557$ for convite). All screens—classical and network-based—are evaluated against this common control on the unified close-margin benchmark.

5.5. Variable definitions

Table 2 summarizes the key variables used in the analysis.

Table 2: Variable Definitions

Variable	Definition
MV_a	Margin of victory: $(b_a^{\text{runner-up}} - b_a^{\text{win}})/b_a^{\text{win}}$
CV_a	Coefficient of variation of firm-best bids in auction a
S_a	Bid spread: $(\max - \min)/\text{mean}$ of firm-best bids
N_a	Number of distinct firms submitting bids in auction a
Incumbent_a	$= 1$ if the winning firm won the preceding auction in the same market
Last bidder_a	$= 1$ if the winning firm submitted the final bid
W_{ij}	Shared workers: count of distinct PIS appearing at both firms i and j
J_{ij}	Jaccard similarity: $W_{ij}/(N_i + N_j - W_{ij})$
C_{ij}	Co-bidding count: number of auctions where both i and j bid

6. Results

We present the empirical results in three parts. Section 6.1 reports the performance of classical bid-distribution screens and the close-margin RD on the unified benchmark. Section 6.2 reports the performance of worker-flow and co-bidding network screens. Section 6.3 reports the specialization-placebo decomposition.

6.1. Classical screens and the close-margin RD

Table 6 presents the unified benchmark comparison of all screens on the same close-margin pregão sample ($|MV| \leq 0.10$). Panel A reports the classical bid-distribution screens on the cartel-active benchmark (N = 2,319 cartel-tainted auctions vs. N = 527,259 clean controls).

All four classical screens fire in the direction predicted by the bid-rotation model: the coefficient of variation ($\Delta = -0.035$, $d = -0.07$), bid spread ($\Delta = -0.173$, $d = -0.10$), and number of active bidders ($\Delta = -0.87$, $d = -0.24$) are compressed at cartel-tainted auctions relative to clean controls. Welch t-statistics are large and statistically significant, owing to the massive control sample. Yet the effect sizes are small: Cohen's d

ranges from -0.07 to -0.24, indicating near-total distributional overlap between cartel and clean auctions. The AUCs reflect this overlap, ranging from 0.52 (coefficient of variation) to 0.56 (number of bidders)—above chance but far too low for investigative triage. A Wilcoxon rank-sum test confirms the weak rank-order separation ($p < 10^{-4}$ for each screen, consistent with the AUC direction).

The margin of victory exhibits an anomalous pattern: $\Delta MV = +0.004$ ($d = +0.17$), with an AUC of 0.46—slightly *below* chance and in the *opposite* direction to the bid-rotation prediction. Cartel-active auctions have marginally *larger* winning margins, not smaller. We interpret this as consistent with a partial phantom-bidding mechanism in which cover bids are inflated relative to the designated winner's price, widening rather than compressing the margin. This finding warrants further investigation but does not affect the headline conclusion: classical screens as a portfolio do not achieve operational discriminative power on this benchmark.

A direct replication of the Kawai et al. (2023) close-margin RD on the cartel-active subsample yields no statistically detectable discontinuity in incumbency ($\hat{\beta} = 0.018$, SE $= 0.005$ on the full sample; $p > 0.15$ on the cartel-active subsample) or last-bid behavior at the cutoff. A density test in the spirit of Imhof et al. (2018) detects no manipulation of the running variable ($T_{\mathrm{CJM}} \approx 0$, $p = 1.0$ on the cartel-active benchmark). Neither the RD design nor the density test flags the cartel-active auctions as anomalous relative to competitive bidding.

6.2. Network-based screens

Panel B of Table 6 reports the network screens on the same unified benchmark ($N = 2{,}319$ cartel-active pregão auctions, $|MV| \leq 0.10$).

The worker-flow screens achieve markedly higher discriminative power. The shared-worker count yields an AUC of 0.750 ($\Delta = +3.70$, $t = +14.1$), and the Jaccard similarity yields an AUC of 0.739 ($\Delta = +0.0014$, $d = +0.19$, $t = +20.7$). Both outperform the best classical screen (number of bidders, AUC $= 0.56$) by roughly 20 percentage points. A third screen constructed from co-bidding density yields an AUC of 0.695, providing independent corroboration through a distinct information channel.

Critically, the worker-flow signal is not an artifact of sectoral composition. Restricting attention to firm pairs that share the same 2-digit CNAE sectoral classification reduces the Jaccard AUC only marginally, from 0.739 to approximately 0.730. The residual of worker-flow after partialling out the log co-bidding density of each pair achieves an AUC

of 0.749—essentially unchanged from the raw baseline, indicating that worker mobility carries detection information above and beyond what co-bidding patterns alone can reveal.

On the convite subsample ($N = 263$ cartel-active, $|MV| \leq 0.10$), the worker-flow screens maintain strong performance: shared-worker AUC = 0.704, Jaccard AUC = 0.701. The consistency of network-screen performance across auction formats—in contrast to the format-sensitive behavior of classical screens—supports the view that worker-flow connectivity captures a channel of coordination that is invariant to the procedural details of the bidding mechanism.

A natural question is what the cartel-active AUC actually measures, given that in most auctions only *one* of the two firms in the (winner, runner-up) pair is a convicted cartelist. Of the 2,319 cartel-active auctions, 996 have the *winner* as the cartel-active firm and the runner-up as a non-cartel firm; 1,313 have the *runner-up* as the cartel-active firm; and only 10 have both firms cartel-active. Decomposing the AUC by cartel role clarifies the interpretation:

Subgroup	N	AUC (shared workers)	AUC (Jaccard)
Winner cartel, runner-up clean	996	0.741	0.730
Runner-up cartel, winner clean	1,313	0.759	0.747
Both cartel-active	10	0.558	0.550

Three observations emerge. *First*, the AUC is remarkably stable across the two one-sided subgroups (0.74–0.76), indicating that the detection signal is symmetric: it does not matter whether the cartel firm wins or loses the auction. This rules out the concern that the screen merely captures a "winner effect" driven by a cartel firm's incumbency or market dominance. *Second*, the signal is present even when the *runner-up* is the cartel firm (AUC = 0.759), confirming that elevated worker-flow connectivity is a property of cartel-firm pairs in thin labor markets, not an artifact of winning per se. *Third*, intra-cartel pairs (both firms cartel-active, $N = 10$) exhibit the highest mean shared-worker count ($\bar{W} = 28.4$, compared with 6.5 for winner-only and 2.1 for clean controls) but an AUC near chance (0.56), reflecting severe power constraints at small N rather than an absence of signal.

The decomposition clarifies the mechanism: the worker-flow screen detects that cartel firms are embedded in thin specialized-labor markets where personnel flows with

all rivals—not just co-conspirators—are dense. The specialization placebo (Section 6.3) further separates this market-embeddedness component from the cartel-specific coordination residual. The headline AUC of 0.75 thus captures both channels; the cartel-specific residual, net of market specialization, is approximately 0.64 (Section 6.3).

6.3. Specialization-placebo decomposition

Table 7 reports the three-way comparison of worker-flow distributions across the cartel, same-market placebo, and clean-global samples (pregão, $|MV| \leq 0.10$).

Cartel pairs share at least one worker in 57.9% of cases, compared with 31.1% for same-market placebo pairs and 9.9% for pairs in other markets. The Jaccard AUC against the global clean baseline is 0.740 for cartel pairs and 0.605 for same-market placebo pairs. In a linear decomposition of the total AUC gain (cartel vs. clean minus 0.5):

- The **specialization component** (placebo vs. clean) absorbs approximately 0.105 AUC points, or roughly 44% of the total signal.
- The **cartel-specific residual** (cartel vs. placebo) accounts for approximately 0.138 AUC points, or roughly 56% of the total signal.

Both components are consistent with the institutional-thinness mechanism of Section 4.6: thin specialized-labor markets generate spontaneous labor circulation among same-market firms (the specialization component), and cartels additionally reinforce those flows to sustain coordination (the cartel-specific residual). The two-component structure also has a practical implication: a worker-flow screen deployed without conditioning on market identity would capture both components, potentially flagging non-colluding same-market firm pairs as suspicious. Conditioning on market or product code absorbs the specialization component, yielding a purer cartel signal at the cost of a modest reduction in overall AUC (from 0.740 to 0.638).

6.4. Heterogeneity by market thickness

The institutional-thinness model of Section 4.6 predicts that worker-flow screens should outperform classical screens most sharply in thin markets, where the cheapest coordination technology operates through personnel flows rather than bid-pattern manipulation. We test this prediction by splitting the cartel-active benchmark according to

market thickness, measured as the median number of bidders per auction for each item code in the clean control sample. Markets with a control-sample median at or below the overall median of 5 firms are classified as "thin"; those above are classified as "thick." This proxy is constructed entirely from the non-cartel sample to avoid contamination by cartel-induced changes in bidder counts.

Table 3 reports the results. The pattern is striking. In thin markets ($N_{\text{cartel}} = 1{,}618$), classical screens fire in the *wrong* direction: CV AUC = 0.45, spread AUC = 0.44, and bidder-count AUC = 0.40—all below chance. Cartel-tainted auctions in thin markets exhibit *more* bid dispersion and *more* active bidders than clean controls, consistent with a phantom-bidding or cover-bid mechanism in which all cartel members participate and submit inflated bids. By contrast, in thick markets ($N_{\text{cartel}} = 416$), classical screens fire in the predicted direction: CV AUC = 0.59, spread = 0.60, bidder count = 0.60—substantively above chance and consistent with the standard bid-rotation model.

Worker-flow screens show the opposite gradient. In thin markets, shared-worker AUC reaches 0.774 and Jaccard AUC reaches 0.758; in thick markets, both decline to approximately 0.70. The gap between the best classical screen and the worker-flow screen is 33 percentage points in thin markets but only 10 percentage points in thick markets—a three-fold difference that confirms the model's core prediction.

The tercile decomposition (reported in the notes to Table 3) reveals additional structure: the worker-flow AUC peaks in mid-thickness markets (tercile 2: AUC = 0.80) rather than in the thinnest markets (tercile 1: AUC = 0.75). This non-monotonicity is consistent with the specialization-placebo finding of Section 6.3: in the very thinnest markets, spontaneous same-market labor circulation is so dense that it partially compresses the cartel-vs-clean AUC from above, even as the cartel-specific signal remains strong.

Table 3: Screen Performance by Market Thickness

	N_{cartel}	N_{clean}	CV	Spread	N_{firms}	Shared workers	Jaccard
Thin (≤ 5 firms)	1,618	217,662	0.447	0.436	0.400	**0.774**	**0.758**
Thick (> 5 firms)	416	224,074	0.594	0.604	0.604	0.699	0.694
All (benchmark)	2,319	527,259	0.523	0.531	0.555	**0.750**	**0.739**

Notes: Market thickness is the median number of bidders per auction for each item code, computed on clean-control auctions with ≥ 5 observations. "Thin" = median ≤ 5; "Thick" = median > 5. Classical screen AUCs use "lower is cartel" convention. Tercile decomposition: T1 (median = 4): $N_t = 1{,}191$, shared-workers AUC = 0.751, CV AUC = 0.382; T2 (median = 5): $N_t = 602$, shared-workers AUC = 0.797, CV AUC = 0.533; T3 (median = 8): $N_t = 241$, shared-workers AUC = 0.644, CV AUC = 0.612.

6.5. Model predictions vs. empirical findings

Table 4 summarizes the correspondence between the three testable predictions of the institutional-thinness model (Section 4.6) and the empirical results documented in this section.

Table 4: Model Predictions and Empirical Results

Prediction	Empirical test	Result
P1. In thin markets, classical AUCs ≈ 0.5; worker-flow AUCs > 0.5.	Table 3: median split by market thickness	Classical AUC = 0.40–0.45 (inverted); worker-flow AUC = 0.77. **Confirmed.**
P2. Same-market placebo exhibits elevated worker-flow relative to random pairs; cartel pairs exhibit still more.	Table 7: three-way decomposition	Specialization component = 44%; cartel residual = 56%. Both present. **Confirmed.**
P3. Classical enforcement under-detects cartels where deterrence value is highest.	Table 3: gap thin vs. thick	Gap = 33 pp in thin markets; 10 pp in thick. **Confirmed.**

All three predictions are borne out in the data. The model's qualitative framework—that the cheapest coordination technology determines which screen family carries the detection signal—is thus not merely an organizing device but a tested empirical proposition, at least within the institutional setting of São Paulo procurement.

6.6. Selection and the interpretation of the benchmark

A natural concern with any conviction-anchored benchmark is that the cartels CADE detected and convicted are not a random sample of all cartels that existed during the study period. If the detection channel itself correlates with worker-flow intensity—for instance, if CADE is more likely to investigate firms when their personnel connections are visible to investigators—then the worker-flow AUC may overstate the screen's performance on the universe of undetected cartels. We address this concern in three ways.

First, we document the evidentiary basis of each of the seven convicted cartels from the public record of CADE's administrative proceedings. Table 5 summarizes the primary detection channel and the nature of the evidence that sustained each conviction.

The evidence channels are heterogeneous: one case (aquecedores solares) was convicted "based exclusively on indirect evidence"—that is, bid-pattern analysis alone; one case (transporte escolar) relied on IP address coincidence during electronic bidding; one case (sacos de lixo) originated from a state-level police operation (*Operação Colludium*); and the remaining four were built from leniency confessions, procurement-document analysis, or investigation-triggered dawn raids. Crucially, *none* of the seven cases mentions matched employer–employee data, RAIS records, or inter-firm personnel connections as part of the evidentiary basis for the investigation or conviction. CADE does not, as a matter of standard practice, use worker-flow analysis as an investigative tool—the data linkage we construct in this paper is, to our knowledge, the first of its kind. The detection channel (CADE investigations driven by complaints, leniency, police operations, and bid-pattern screens) is therefore independent of the screening channel (worker-flow connectivity measured from RAIS).

Second, we note that the worker-flow screen is constructed from an entirely external administrative data source (RAIS, held by the Ministry of Labor) that is not part of CADE's standard enforcement toolkit. Unlike co-bidding patterns—which are constructed from the same procurement data that CADE analysts routinely examine—worker-flow connectivity requires a separate data-access agreement and a non-trivial linkage procedure. The informational independence of the two channels reduces the scope for mechanical circularity.

Third, we observe that the worker-flow AUC is relatively stable across the seven cartels despite the heterogeneity in detection channels. If selection-on-detection were driving the result, we would expect the worker-flow signal to be concentrated in cases where personnel connections played a role in the investigation. Instead, the signal is present even in the aquecedores solares case—convicted on bid-pattern evidence alone—and in the transporte escolar case—detected through IP forensics. While this cross-cartel stability is suggestive rather than definitive (small N of cartels limits formal testing), it is inconsistent with a pure selection story.

These observations do not fully eliminate the selection concern. It remains possible that firms with dense worker-flow networks are easier to detect through *any* channel—for instance, because the same organizational proximity that generates personnel flows also generates careless document trails or complaints from disgruntled employees. A definitive test would require evaluating the worker-flow screen against cartels detected through channels that are *provably* orthogonal to personnel connectivity, such as pure

leniency applications from firms with no prior labor-market overlap. We leave this test to future work and flag the selection caveat as the principal interpretive limitation of our benchmark design.

Table 5: Evidence Channels in the Seven Convicted Cartels

Cartel (sector)	CADE process	Period	Primary evidence channel
Merenda escolar	08012.010022/ 2008-16	2006–2013	Procurement documents (40,000+ docs analyzed); geographic market division
Trens e metrôs	08700.004617/ 2013-41	1999–2013	Leniency and investigation; subcontracting arrangements between winners and losers
Aquecedores solares	08012.001273/ 2010-24	2009–2010	Indirect evidence only (bid-pattern analysis); no direct evidence of agreement
Transporte escolar	08700.005876/ 2019-85	2019	IP address coincidence during electronic bidding
Cafeterias Infraero	08700.007278/ 2015-17	2014	Investigation of in-person tender procedures
Sacos de lixo	08700.005789/ 2015-02	2008–2014	Police operation (*Op. Colludium*, MP Bauru); dawn raids and document seizures
Medicamentos	08012.002222/ 2011-09	2007–2011	Hub-and-spoke investigation; leniency and administrative proceedings

Notes: Evidence channels documented from CADE public press releases and Tribunal decisions (see Appendix for full citations). None of the seven cases lists matched employer–employee data, RAIS records, or inter-firm worker flows as part of the evidentiary basis.

7. Discussion and Conclusion

This paper assembles the first conviction-anchored cartel-detection benchmark that integrates auction-level bid data, matched employer–employee records, and formal antitrust rulings. The benchmark links seven CADE-convicted procurement cartels in São Paulo to 3.8 million bids on the state's electronic procurement platform and to the universe of formal-sector employment relationships in RAIS. On a unified close-margin sample of roughly 2,300 pregão auctions, we document three main findings.

Table 6: Unified Benchmark: All Screens on the Same Close-Margin Sample

	Pregão ($N_t = 2{,}319$)		Convite ($N_t = 263$)	
Screen	Cohen's d	AUC	Cohen's d	AUC
Panel A: Classical bid-distribution screens				
Coefficient of variation (CV)	−0.07	0.523	−0.01	0.515
Bid spread (S)	−0.10	0.531	+0.04	0.502
Margin of victory (MV)	+0.17	0.457	+0.14	0.456
Number of bidders (N)	−0.24	0.555	+0.06	0.467
Panel B: Network-based screens				
Shared workers (W_{ij})	+0.06	**0.750**	+0.47	**0.704**
Jaccard similarity (J_{ij})	+0.19	**0.739**	+0.09	**0.701**
Co-bidding density (C_{ij})	+0.14	0.695	−0.04	0.525

Notes: All screens evaluated on the unified close-margin benchmark ($|MV| \leq 0.10$) against a common clean control ($N_0 = 527{,}259$ pregão; $N_0 = 519{,}557$ convite). Classical screen AUCs use the "lower is cartel" convention; values above 0.5 indicate detection in the predicted direction. Network screen AUCs use "higher is cartel." Bold indicates AUC ≥ 0.70. Cohen's d is the standardized mean difference (pooled SD). Source: script 23.

Table 7: Specialization-Placebo Decomposition of Worker-Flow Signal

Sample	N	% with ≥ 1 shared worker	Mean Jaccard	AUC vs. clean
Cartel-active	2,321	57.9%	0.0022	0.740
Placebo (same market)	5,313	31.1%	0.0012	0.605
Clean (global)	523,414	9.9%	0.0008	—
Signal decomposition (Jaccard AUC relative to clean baseline)				
Total signal (cartel vs. clean − 0.5)				+0.240
Specialization component (placebo vs. clean)				+0.105 (44%)
Cartel-specific residual (cartel vs. placebo)				+0.138 (56%)

Notes: Pregão, $|MV| \leq 0.10$. Placebo consists of non-cartel firm pairs bidding on the same BEC item code in the same year as each cartel-active auction. AUC computed via Mann–Whitney U (Jaccard similarity, higher is cartel). Percentages in decomposition are shares of total signal. Source: v3 specialization placebo.

First, canonical variance-based screens fire in the direction predicted by the bid-rotation model—bid dispersion, coefficient of variation, and bidder counts are weakly compressed at cartel-tainted auctions—but achieve AUCs of only 0.52–0.56, far too low for investigative triage. The close-margin RD of Kawai et al. (2023) likewise produces a null on the cartel-active subsample. Neither result repudiates the underlying designs; both warn that screens validated in thick-labor environments lose operational discriminating power when redeployed to settings where cartel coordination relies on alternative technologies.

Second, network-based screens constructed from inter-firm worker flows achieve AUCs of 0.75 (shared workers) and 0.74 (Jaccard similarity), outperforming the best classical screen by roughly 20 percentage points. A co-bidding density screen yields 0.70. The worker-flow signal survives conditioning on sector and on co-bidding intensity, indicating that it carries detection information beyond what bid-pattern or co-participation data alone can reveal.

Third, a specialization placebo decomposes the worker-flow signal into a market-specialization component (approximately 44% of the total) and a cartel-specific coordination residual (approximately 56%). Both components are consistent with an institutional-thinness mechanism: thin specialized-labor markets generate spontaneous inter-firm labor circulation, and cartels reinforce those flows to sustain coordination. This two-component structure implies that conditioning worker-flow screens on product-market identity sharpens the cartel signal at a modest cost in overall AUC.

Policy implications. For competition authorities in middle-income economies with access to employer–employee microdata—increasingly common as countries digitize social-security and labor records—worker-flow screens offer a deployable complement to canonical bid-pattern tests. The marginal cost of constructing such screens is negligible once the data linkage exists. Our results suggest that enforcement strategies relying exclusively on classical screens may systematically under-detect cartels in precisely the institutional environments where deterrence capacity is weakest.

Limitations and directions for future work. The principal interpretive limitation is selection-on-detection: CADE convictions are not a random sample of all cartels. As documented in Section 6.6, none of the seven convicted cartels was detected through worker-flow or employer–employee evidence—CADE relied on procurement documents, leniency confessions, police operations, and (in one case) IP-address forensics—but we

cannot fully rule out that firms with dense personnel networks are easier to detect through *any* channel. Future work should evaluate the screen against cartels detected through provably orthogonal channels. The strict pair-matched sample ($N = 10$) is too small to draw conclusions about head-to-head cartel coordination; larger ground-truth sets, potentially assembled from cross-state CADE linkages, would allow sharper tests. Finally, the simple model of Section 4.6 treats bid-pattern and worker-flow coordination as substitutes; a richer framework allowing complementarity could generate additional testable predictions.

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A. Appendix

A.1. Bandwidth sensitivity

Table A.1 reports AUCs for all screens across six bandwidth choices for the close-margin envelope. The benchmark bandwidth of $|MV| \leq 0.10$ used throughout the paper is neither the most nor the least favorable for the worker-flow screens. At narrower bandwidths (0.02, 0.05), worker-flow AUCs are marginally higher (0.76, 0.75) but sample sizes are smaller; at wider bandwidths (0.20, 0.50), AUCs decline modestly (0.74, 0.72) as the sample dilutes toward the full population. Classical screen AUCs follow the same monotonic pattern. The gap between the best classical screen (number of bidders) and the worker-flow screens remains stable at approximately 18–20 percentage points across all bandwidths.

Table A.1: AUC Sensitivity to Close-Margin Bandwidth (Pregão, Cartel-Active)

$\|MV\| <$	N_{cartel}	N_{clean}	CV	Spread	N_{firms}	Shared workers	Jaccard
0.02	989	277,038	0.546	0.558	0.580	0.758	0.747
0.05	1,700	416,582	0.539	0.548	0.565	0.760	0.749
0.10	**2,319**	**527,259**	**0.523**	**0.531**	**0.555**	**0.750**	**0.739**
0.15	2,607	573,196	0.518	0.526	0.554	0.745	0.734
0.20	2,784	602,883	0.517	0.525	0.555	0.743	0.732
0.50	3,277	693,331	0.518	0.526	0.552	0.735	0.723

Notes: Each row reports AUCs from the unified benchmark (script 23) at the indicated bandwidth. Classical screen AUCs use the "lower is cartel" convention (negated scores). Bold row indicates the benchmark bandwidth used in the main analysis. All AUCs computed via Mann–Whitney U with average-rank tie-handling.